Current activities of the ELIXIR Machine Learning Focus Group

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Motivations Machine Learning (ML) has emerged as a discipline that enables computers to assist humans in making sense of large and complex data sets[1,2]. With the drop in the cost of high-throughput technologies, large amounts of omics data are being generated and made accessible to researchers[3-5]. Analyzing these complex high-volume data is not trivial, and the use of classical statistics cannot explore their full potential. Machine Learning can thus be very useful in mining large omics datasets to uncover new insights that can consequently lead to the advancement of Life Sciences.

Aims The main aims of ELIXIR Machine Learning Focus Group (MLFG) is to increase the reproducibility and transparency of ML methods for researchers, the journal reviewers and the wider community. One of the most pressing issues is to agree to a standardized data structure to describe the most relevant features of ML studies being published. The development of standardized reporting model has the potential to make a major impact in increasing the quality of ML-based publications.

Results Our focus group was initiated in October 2019, in order to capture the emerging need in Machine Learning expertise across the network. The main result so far is the DOME Recommendations, a set of community-wide recommendations for reporting supervised machine learning–based analyses applied to biological studies.

Tasks Currently the ELIXIR Machine Learning Focus Group is working on three main tasks:

(1) using the DOME recommendations to annotate relevant literature in order to gain insights into the level of adherence to DOME;

(2) evaluating the gold standard datasets widely used in ML process in order to define and describe the aspects of a gold standard with particular focus on human data;

(3) reviewing the efforts around synthetic data in order to establish a set of best practices for their use and application in ML.

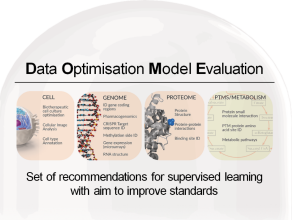
DOME Recommendations

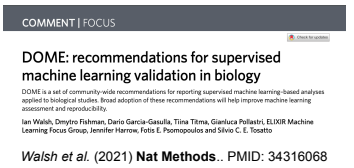
Data: Preprocessing data properly, and using it in a knowledgeable manner is the only way to obtain good generalization.

Optimization: Problems associated with a poor choice of optimization strategy.

Model: Fallacies of important aspects of the ML models.

Evaluation: Valid assessment methodology for any final model with adequate and comprehensive measures.




COMMENT | FOCUS

DOME: recommendations for supervised machine learning validation in biology

DOME is a set of community-wide recommendations for reporting supervised machine learning-based analyses applied to biological studies. Broad adoption of these recommendations will help improve machine-learning assessment and reproducibility.

Ian Walsh, Dmytro Fishman, Dario Garcia-Gasulla, Tiina Titma, Gianluca Pollastri, ELIXIR Machine Learning Focus Group, Jennifer Harrow, Fotis E. Psomopoulos and Silvio C. E. Tosatto

Walsh et al. (2021) **Nat Methods**.. PMID: 34316068

Main Aims

- **Standards** for Machine Learning
- **Benchmarking** of Machine Learning tools
- Machine Learning **FAIR/ reproducibility**
- **Integration** across ELIXIR Communities
- **Training** for Machine Learning

ML Focus Group Tasks

TASK 1:

- Initiate work on building a corpus of papers annotated following DOME → Crowdsource annotation effort within the Focus Group.
- Generate a registry of annotated papers.
- Define a controlled vocabulary for DOME annotation, as the groundwork towards a machine-actionable/machine-readable version of DOME.

TASK 2:

- Selection of papers reporting datasets for training supervised machine learning methods with specific focus on omics data on human.
- Review and annotation of the collected datasets.
- Building *a* repository with curated benchmark datasets.

TASK 3:

- Collection, classification and curation of available datasets of synthetic data.
- Review of the available tools for synthetic data generation.
- Generation of an Infrastructure for synthetic health data
- Align the description of the synthetic datasets to the DOME recommendations

Info

web: https://dome-ml.org

References

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MLFG Tasks Participants

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